COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:		**SECTION:**	
DATE PERFORMED:		**DATE SUBMITTED:**	
NAME:		**INSTRUCTOR:**	

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

I. Identify the Requirements

Linux distribution and version used

Linux Distribution: Ubuntu Server 24.04 LTS

Ubuntu is used for this research because Nagios Core officially lists Ubuntu among its supported Linux distributions, including Ubuntu 24.04 LTS.

Step #	Command / Action	Purpose	Expected Results
1	sudo apt update	Updates the package repository.	Package list is updated.
2	sudo apt install apache2 php build-essential libgd-dev libssl-dev wget unzip apache2-utils	Installs required packages and dependencies.	Dependencies are installed.
3	Create nagios user/group and nagcmd group.	Provides the accounts and permissions needed by Nagios.	Required users/groups exist.
4	Add nagios and www-data to nagcmd.	Allows Nagios and Apache to use the command interface.	Users have the required permissions.
5	Download Nagios Core source archive.	Gets the Nagios source code.	Source archive is downloaded.
6	tar xzf nagios-*.tar.gz	Extracts the source files.	Nagios source directory is created.

7	./configure --with-command-group=nagcmd	Prepares Nagios for compilation.	Makefiles are generated.
8	make all	Compiles Nagios Core.	Nagios is successfully built.
9	sudo make install	Installs Nagios Core.	Nagios files are installed.
10	sudo make install-init and sudo make install-config	Installs service and basic configuration files.	Nagios service/configuration is available.
11	sudo make install-commandmode	Configures external command permissions.	Command interface is configured.
12	sudo make install-webconf	Installs Apache configuration for Nagios.	Nagios web interface is configured.
13	sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin	Creates web authentication credentials.	nagiosadmin login is created.
14	Download and extract Nagios Plugins.	Gets the plugins used for monitoring.	Plugin source files are available.
15	./configure → make → sudo make install	Builds and installs Nagios Plugins.	Monitoring plugins are installed.
16	sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg	Checks the configuration for errors.	Configuration passes validation.
17	sudo systemctl enable --now apache2 nagios	Enables and starts Apache and Nagios.	Both services are running.
18	Open http://<server-ip>/nagios/	Accesses the Nagios web interface.	Nagios login page appears.
19	Log in and check Hosts/Services.	Verifies that Nagios is working.	Monitoring information is displayed.

III. Identify What Ansible Should Automate

Manual Installation Step	Can Ansible Automate It?	Possible Ansible Module
1. Install required packages	Yes	ansible.builtin.apt
2. Create Nagios user/group	Yes	ansible.builtin.user / group
3. Create nagcmd group	Yes	ansible.builtin.group
4. Add users to groups	Yes	ansible.builtin.user
5. Download Nagios source	Yes	ansible.builtin.get_url
6. Extract source files	Yes	ansible.builtin.unarchive
7. Run ./configure	Yes	ansible.builtin.command
8. Compile Nagios	Yes	ansible.builtin.command
9. Install Nagios	Yes	ansible.builtin.command
10. Configure Nagios files	Yes	ansible.builtin.template
11. Install Nagios Plugins	Yes	ansible.builtin.command
12. Configure Apache	Yes	ansible.builtin.template
13. Create web authentication	Yes	community.general.htpasswd
14. Validate Nagios configuration	Yes	ansible.builtin.command
15. Enable and start services	Yes	ansible.builtin.systemd
16. Check service status	Yes	ansible.builtin.systemd
17. Open Nagios in browser	No / Manual	—

IV. Preparation for the Hands-on Activity 8

Manual	Ansible
1. sudo apt update	ansible.builtin.apt
2. sudo apt install apache2 php ...	ansible.builtin.apt
3. sudo useradd -m nagios	ansible.builtin.user
4. sudo groupadd nagios	ansible.builtin.group
5. sudo groupadd nagcmd	ansible.builtin.group
6. sudo usermod -a -G nagcmd nagios	ansible.builtin.user
7. wget https://.../nagios-4.x.x.tar.gz	ansible.builtin.get_url
8. tar xzf nagios-4.x.x.tar.gz	ansible.builtin.unarchive
9. ./configure --with-command-group=nagcmd	ansible.builtin.command
10. make all	ansible.builtin.command
11. sudo make install	ansible.builtin.command
12. sudo make install-config	ansible.builtin.command
13. sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin	community.general.htpasswd
14. sudo systemctl enable --now apache2 nagios	ansible.builtin.systemd
15. sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg	ansible.builtin.command

Research Questions

Answer the following questions briefly.

1. What is Nagios Core?

Nagios Core is an open-source monitoring system used to monitor servers, computers, networks, and services.

2. What is the purpose of Nagios plugins?

Nagios plugins are used to check if a service or system is working properly, like checking CPU, disk space, HTTP, or SSH.

3. Why does Nagios require additional packages or dependencies before installation?

Nagios needs additional packages because some tools and libraries are needed to build, install, and run Nagios properly.

4. What is the purpose of the Nagios user and group?

The Nagios user and group are used to run Nagios with the correct permissions instead of using the root account.

5. What is the purpose of the web server in a Nagios installation?

The web server, such as Apache, is used to access and view the Nagios monitoring interface through a web browser.

6. Why is configuration validation necessary before starting Nagios?

It is important to check the configuration first so we can find errors before starting the Nagios service.

7. What happens if the Nagios configuration contains an error?

Nagios may fail to start or the monitoring system may not work properly.

8. Which installation steps require elevated/root privileges?

Installing packages, creating system users and groups, installing Nagios files, configuring Apache, and starting or enabling services usually require sudo or root privileges.

9. Which steps are idempotent when implemented in Ansible?

Installing packages, creating users and groups, creating directories, managing configuration files, and starting/enabling services can be idempotent when using the proper Ansible modules.

10. Which manual installation steps do you think will be the most difficult to automate using Ansible? Explain why.

I think compiling and installing Nagios from source will be the most difficult because it uses commands like ./configure, make, and make install. These commands need to be handled carefully in Ansible to make sure they do not run unnecessarily every time.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.